

Computer vision task 1 exercise 5 (page 1)

17a) we have:
$$\left. \begin{aligned} v' &= x_2' - x_1' \\ v &= x_2 - x_1 \end{aligned} \right\}$$

And: $x' = sRx + t$

$$\Rightarrow v' = (sRx_2 + t) - (sRx_1 + t)$$

$$\Rightarrow v' = sR(x_2 - x_1) = sRv$$

$$\Rightarrow \|v'\| = \|sRv\| = s\|Rv\|$$

But: $\|Rv\| = \|v\|$ since R rotation doesn't change length

$$\Rightarrow \|v'\| = s\|v\| \Rightarrow s = \frac{\|v'\|}{\|v\|}$$

17b) Let $u = \frac{v}{\|v\|}$ and $u' = \frac{v'}{\|v'\|}$

we have: $u' = Ru \Rightarrow \begin{pmatrix} u'_x \\ u'_y \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} u_x \\ u_y \end{pmatrix}$

$$\Rightarrow \begin{cases} u'_x = \cos \theta u_x - \sin \theta u_y \\ u'_y = \sin \theta u_x + \cos \theta u_y \end{cases}$$

$$\Rightarrow \begin{cases} \cos \theta = u'_x u_x + u'_y u_y \\ \sin \theta = u'_y u_x - u'_x u_y \end{cases} \Rightarrow \begin{cases} \theta = \arccos(u'_x u_x + u'_y u_y) \\ \theta = \arcsin(u'_y u_x - u'_x u_y) \end{cases}$$

1c) $x_1' = sRx_1 + t \Rightarrow t = x_1' - sRx_1$

1d) we have:

$$\begin{cases} x_1 = (\sqrt{3}, 1) \rightarrow x_1' = (0, 0) \\ x_2 = (0, 3) \rightarrow x_2' = (9, \sqrt{3}) \end{cases}$$

$$\Rightarrow v = x_2 - x_1 = (-\sqrt{3}, 2) \Rightarrow \|v\| = \sqrt{3+4} = \sqrt{7}$$

$$v' = x_2' - x_1' = (9, \sqrt{3}) \Rightarrow \|v'\| = \sqrt{81+3} = \sqrt{84}$$

$$\Rightarrow s = \frac{\|v'\|}{\|v\|} \text{ (from task a)} = \frac{\sqrt{84}}{\sqrt{7}} = 2\sqrt{3}$$

we have: $\frac{v}{\|v\|} \cdot \frac{v'}{\|v'\|} = \frac{vv'}{\|v\|\|v'\|} = \cos(\vec{v}, \vec{v}') = \cos \theta$

$$\Rightarrow \frac{-9\sqrt{3} + 2\sqrt{3}}{\sqrt{7} \cdot \sqrt{84}} = \cos \theta \Rightarrow \cos \theta = -\frac{1}{2} \Rightarrow \theta = 120^\circ = \frac{2\pi}{3} \text{ (rad)}$$

computer union ex 5 (page 2).

$$\theta = \frac{2\pi}{3} \Rightarrow \sin \theta = \sqrt{3}/2.$$

$$\Rightarrow R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}$$

we have from task (c): $t = x_1' - R x_1$

$$\Rightarrow t = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 2\sqrt{3} \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix} \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix}$$

$$\Rightarrow t = -2\sqrt{3} \begin{pmatrix} -\sqrt{3} \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -2\sqrt{3} \end{pmatrix}$$

$$\Rightarrow t = (6, -2\sqrt{3})$$